Benchmarking Large Foundation EEG Models: Current Performance and Future Directions

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***Abstract*—Large EEG foundation models (LEMs) offer superior downstream performance, yet existing evaluations lack consistency in preprocessing and fail to explain the factors driving model success. We present a rigorous empirical study of seven models (LaBraM, CBraMod, BIOT, EEGPT, NeuroLM, NeuroGPT, ST-Transformer) across five benchmark tasks with controlled preprocessing and exhaustive hyperparameter search. Our analysis reveals three key architectural factors determining performance. First, patch encoder design significantly influences task characteristics: convolution-based encoders excel on localized, short-window tasks (e.g., CBraMod achieves 99% accuracy on movement classification), whereas linear encoders enable stronger long-range dependency modeling (EEGPT achieves 83% on sleep staging). Second, pretraining channel diversity critically affects robustness. When downstream tasks use electrode montages that are significantly different from those used during pretraining, performance variance increases by up to 50 times. Third, models pretrained on diverse task distributions generalize more effectively; LaBraM demonstrates strong performance across four of five downstream tasks. These findings provide an evidence-based foundation for architectural design decisions in next-generation EEG foundation models. Code and benchmarks are publicly available for reproducibility.The code is publicly available at:https://github.com/WubbzyFromWuzzleburg/EEG-Review-Paper-Repo/tree/main.**

I. Introduction

Electroencephalography (EEG) is a widely used non-invasive technique for recording brain activity with millisecond temporal resolution. However, EEG signals suffer from low signal-to-noise ratios and high inter-session and inter-subject variability. Moreover, the labor-intensive and expensive process of labeling EEG data limits the size of available supervised datasets. Traditional supervised models trained on such small, task-specific datasets are prone to overfitting and exhibit poor generalization across subjects and tasks, as shown in Figure 1. These challenges have long impeded the development of robust EEG decoding systems and motivated exploration of new methods that can leverage unlabeled data and transfer generalizable features across domains.

Large-scale foundation models have transformed computer vision and natural language processing through pretraining on broad datasets and fine-tuning on diverse downstream tasks. This paradigm has been extended to EEG analysis through Large EEG Models (LEMs)—foundation models that leverage self-supervised learning on massive EEG datasets to extract generalizable neural representations. The standard training procedure consists of two stages as demonstrated in Figure 2: (1) self-supervised pretraining on unlabeled EEG data, where models learn representations by reconstructing masked time-series or spectrogram patches, and (2) supervised fine-tuning on task-specific labeled data. This two-stage approach enables LEMs to distill knowledge from thousands of hours of EEG recordings spanning diverse subjects and experimental conditions, then adapt to new tasks with minimal additional supervision. For example, LaBraM was pretrained on over 2,500 hours of data from approximately 20 datasets using a neural tokenizer and masked reconstruction objective [1]. Rapidly following this paradigm, researchers have developed numerous LEM architectures that achieve state-of-the-art performance on established EEG benchmarks, frequently surpassing conventional task-specific classifiers across clinical diagnosis, affective computing, and brain-computer interfaces [2], [3].

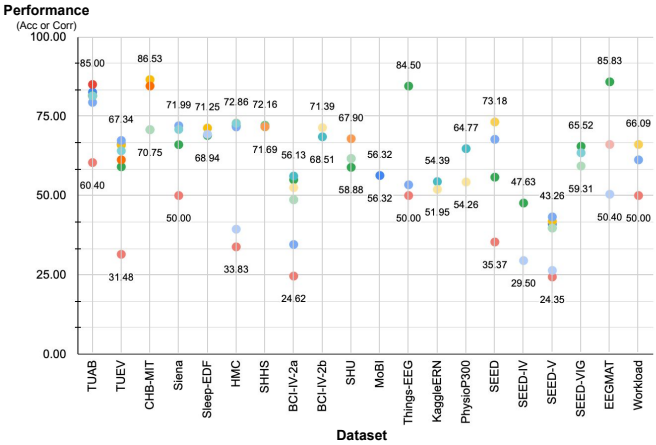

Fig. 3. Performance variability of LaBraM across 20 EEG datasets reported in 15 independent studies. Each point represents the reported performance of LaBraM on a specific dataset in one study, with point colors distinguishing different papers. The x-axis lists the datasets and the y-axis shows performance (%), measured primarily by balanced accuracy (except for MOBI and SEED-VIG, which use Pearson correlation). For each dataset, the highest and lowest reported values are annotated. The figure demonstrates substantial variability in reported performance across studies, reflecting differences in evaluation protocols and experimental settings.

Despite this rapid growth, evaluation practices in the literature remain fragmented and inconsistent, complicating objective comparison between models. Studies typically fine-tune LEMs on different datasets and tasks using study-specific preprocessing pipelines, data splits, and evaluation protocols. Even when the same dataset is used, variations in preprocessing, augmentation, and train-test partitioning can substantially affect reported performance, making it unclear whether observed differences reflect true model superiority or merely methodological artifacts. Figure 3 illustrates this problem by aggregating reported performance of a single model (LaBraM) across 15 independent studies [1], [3]–[16]. Each point represents LaBraM's accuracy (for classification) or Pearson correlation coefficient (for regression) on a given dataset from one paper, with colors distinguishing different studies. The substantial variance in reported performance for the same dataset—with gaps often exceeding 10

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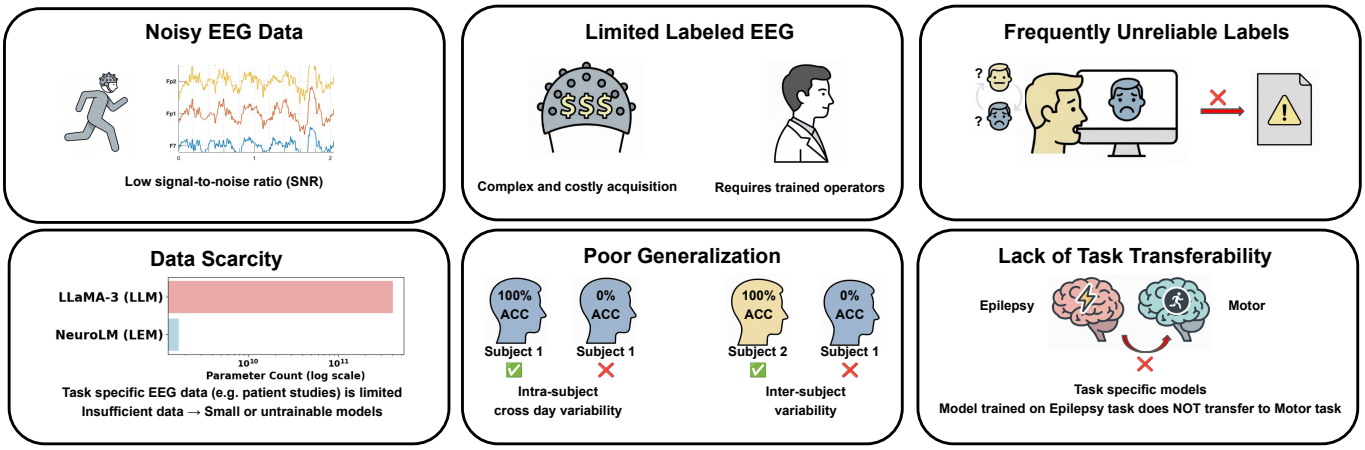

Fig. 1. **Challenges in traditional EEG modeling.** Major obstacles include inherent signal noise, data scarcity due to costly and unreliable labeling, and poor generalization across subjects and tasks. These limitations motivate the development of large foundation models.

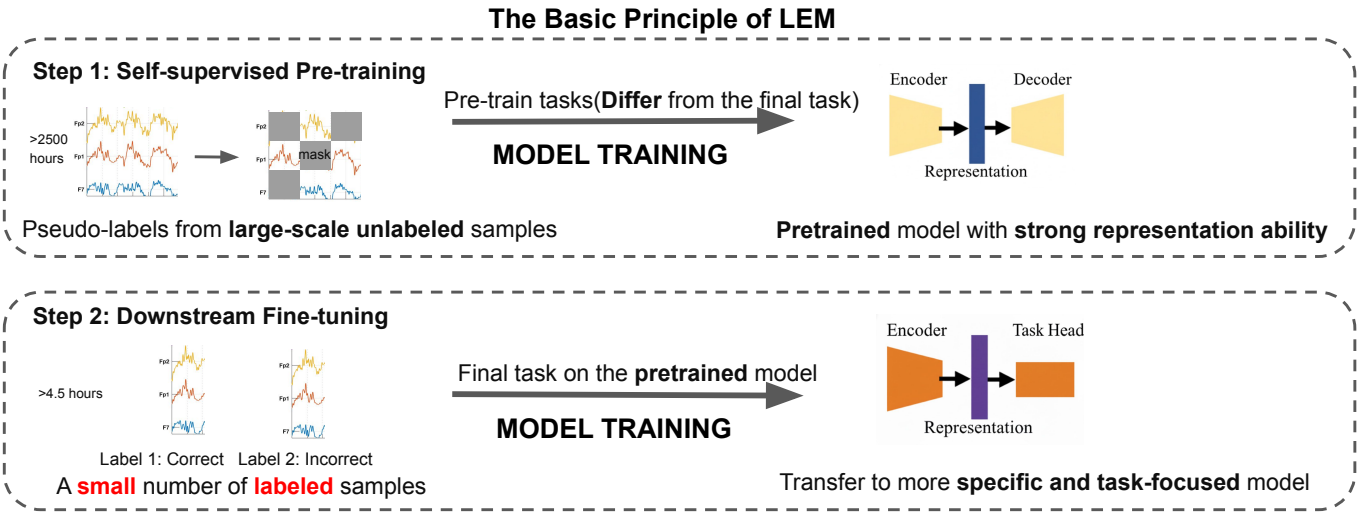

Fig. 2. **Two-stage training pipeline of a Large EEG Model (LEM).** (Stage 1) Self-supervised Pretraining: The model is first trained on extensive unlabeled EEG data using a self-supervised objective (e.g., predicting masked segments or frequency content). This pretraining learns robust cross-subject representations of EEG that capture fundamental spatiotemporal patterns. (Stage 2) Downstream Fine-Tuning: The pretrained model is then fine-tuned on a specific downstream task using a small labeled dataset. Model weights are adjusted with reference to task-specific head to optimize performance for that classification or regression target. By leveraging the robust feature extraction capabilities developed during pretraining, the model can achieve high accuracy with limited task-specific data.

percentage points—demonstrates that evaluation methodology significantly impacts results and undermines the ability to fairly compare models.

While several benchmarks have been recently proposed for large EEG foundation models, many focus exclusively on highly specific application scenarios. For instance, EEG-Bench [17] is narrowly tailored to clinical diagnostics, evaluating 14 datasets across 11 specific diagnostic tasks such as Parkinson's disease, schizophrenia, and epilepsy.

In contrast, the present study provides a broader and more comprehensive evaluation by covering five widely studied downstream tasks: stress detection, movement classification, error-related negativity (ERN), emotion recognition, and sleep staging. This selection uniquely spans diverse temporal scales and spatial patterns. Furthermore, building upon this broad scope, existing benchmarks predominantly rely on highly controlled laboratory or clinical datasets, which fail to reflect the true complexities of real-world applications. To achieve higher ecological validity, our benchmark deliberately incorporates real-world datasets for the stress [18] and movement classification tasks, utilizing data collected during actual classroom learning and unconstrained physical activities. By exposing models to naturalistic noise, severe class imbalances, and motion artifacts, we effectively demonstrate whether these foundation models can maintain their promised robustness and deliver genuine value for practical brain-computer interface (BCI) deployment.

Although some existing benchmarks cover diverse scenarios, they fail to conduct comprehensive performance testing, often relying on fixed or minimally tuned hyperparameters. For example, Brain4FMs [22] employs a fixed learning rate of 1×10^{-5} for the backbone and 1×10^{-4} for the classification head across all models; AdaBrain-Bench [3] merely searches across five learning rate values while keeping the batch size (64) and weight decay (0.05) constant; and EEG-FM-Bench [23] applies the maximum learning rates from open-source implementations while strictly fixing the batch size at 128 and

TABLE I
SUMMARY OF DOWNSTREAM EVALUATION DATASETS

Dataset	Evaluation Unit	Category	Channels	Subjects	Samples	Duration	Classes	Total Length
KaggleERN [19]	Segment	P300 / Error Potential	56	26	5,440	3 s	2 (Error vs Correct)	4.53 h
Self-collected real world movement dataset	Segment	Movement Classification	20	19	40,999	1 s	6 (Motion)	11.39 h
Real world stress dataset [18]	Segment	Stress Classification	30	18	8,334	5 s	2 (Stress Level)	11.58 h
SEED-VII [20]	Video (Voting)	Emotion Classification	62	20	31,124	4 s	6 (Emotions)	34.58 h
ISRUC [21]	Segment	Sleep Staging	6	10	8,889	30 s	5 (Sleep Stages)	74.08 h

TABLE II
TASK-SPECIFIC MODELING REQUIREMENTS INFERRED FROM CLASSICAL EEG ANALYSES.

Task	Temporal Scope	Spatial Characteristics	Time-domain Features	Frequency-domain / Hybrid Characteristics
KaggleERN	Very **short** event-locked window (∼50–100 ms after error); **localized** transient response	Primarily **fronto-central region** with peak at FCz; emphasis on specific electrodes rather than distributed networks	ERP-aligned sharp negative deflection (ERN); precise temporal alignment to error onset	Not emphasized
Stress	Continuous and slowly varying over time; requires tracking **long** nonstationary dynamics rather than fixed short windows	**Distributed** involvement of frontal, temporal, and parietal **regions**; multiple channels contribute concurrently	Not emphasized	Band power changes in delta, theta, alpha, and beta; wavelet-based time–frequency energy capturing evolving spectral patterns
Movement	Event-related dynamics around movement onset and offset; **short** to intermediate windows centered on motor events	Contralateral sensorimotor cortex (C3/C4 dominant); **localized** motor-area channels are primary	ERD during movement and ERS after movement termination	Mu and beta rhythms with joint time–frequency modulation
SEED-VII (Emotion)	**Short** ∼400–600 ms latency ERP responses combined with ongoing emotional state	Frontal hemispheric asymmetry plus parietal/occipital involvement; comparison **across hemispheres and regions**	Late Positive Potential (LPP) amplitude variations	Frontal alpha asymmetry together with ERP components; combined temporal and spectral descriptors
ISRUC (Sleep)	**Integration** of transient local events with long-range temporal context	**Multi-channel** spatial coherence; locally generated events propagating across cortical regions; distributed cortical activation	Transient waveform events	Mixed-frequency activity

weight decay at 0.01.

While this one-size-fits-all configuration appears fair, it is inherently biased because the performance of certain models, such as CBraMod [16], is extremely sensitive to hyperparameter choices as shown in Table IX. A configuration that works well for one model may severely degrade another. A truly fair method should maximize each model's potential to accurately reflect its capability on specific tasks. Our benchmark conducts an exhaustive hyperparameter grid search—varying learning rates, weight decays, and batch sizes to yield 18 to 36 configurations per model-task combination, totaling over 700 experiments as shown in Table III. This ensures that the reported results truly reflect the model's capabilities across domains.

Furthermore, while some existing benchmarks assess models using linear probing (freezing the pretrained backbone), these analyses often yield superficial or misleadingly low estimates of model capability. For instance, EEG-FM-Bench reports a 'catastrophic performance collapse' across models under linear probing, concluding that current pretrained representations lack out-of-the-box utility. However, our benchmark reveals a far more nuanced reality. By conducting rigorous linear probing evaluations, we demonstrate that performance collapse is not universal but strictly tied to architectural design choices.

A second frequently overlooked critical factor is data preprocessing. Current benchmarks uniformly apply standardized preprocessing pipelines. This standardized, one-size-fits all approach to preprocessing ignores the fact that foundation models are coupled with specific preprocessing protocols (e.g., specific filtering frequency bands, reference methods) for pretraining. Specifically, AdaBrain-Bench [3] applies a universal 0.1–75 Hz bandpass filter and site-specific notch filtering based on Fast Fourier Transform, aligns the resampling rate to the target model, and applies dataset-dependent normalization (Z-score for most, 95-percentile for outlier-prone sets like SHU, and unit rescale for Things-EEG). EEG-FM-Bench [23] utilizes a rigid sequential pipeline: 0.1–128 Hz bandpass filter, and 50/60 Hz notch filtering, uniformly downsampling to 256 Hz. Brain4FMs [22] executes a pipeline consisting of downsampling, universal 50/60 Hz notch and bandpass filtering and per-channel Z-score normalization. Similarly, EEG-Bench [17] applies a 0.1–75.0 Hz bandpass filter and notch filtering, resamples all signals to 200 Hz. Our benchmark strictly matches the tailored preprocessing pipeline to each model's original pretraining data distribution (as detailed in Table IV). This approach minimizes domain shift and more accurately reflects the models' true representational capabilities rather than artificial performance degradation caused by mismatched preprocessing parameters.

This study conducts a systematic benchmark of seven representative large EEG foundation models using the tailored preprocessing and hyperparameter tuning approaches described above (LaBraM [1], CBraMod [16], BIOT [24], EEGPT [7], NeuroLM [9], NeuroGPT [25], and ST-Transformer [26]). These models are commonly employed in contemporary EEG analysis and have demonstrated strong empirical performance.

II. BENCHMARK DATASETS AND TASK FORMULATION

To comprehensively evaluate the representation capabilities of foundation models across diverse EEG applications, we benchmarked models on five downstream tasks: error-related negativity (ERN) detection, cognitive load assessment, movement classification, emotion recognition, and sleep stage scoring. These tasks encompass diverse temporal, spatial, and spatiotemporal modeling requirements, thereby providing comprehensive coverage of distinct EEG decoding paradigms (Table I). The detailed modeling requirements and data characteristics for each task are summarized in Table II.

A. KaggleERN: Error-Related Negativity Detection

The KaggleERN dataset employs a P300 Speller paradigm where subjects focus on target characters within a 6×6 alphanumeric matrix. The target characters flash in a pseudo random sequence, acting as rare stimuli that elicit a P300 response, which the system uses to infer and output the attended character. However, the system's predictions are prone to errors. This dataset captures the subjects' EEG responses immediately following the presentation of the system's output to detect error-related potentials (ERP). For our downstream task, we extract 3-second EEG segments following the feedback presentation to perform a binary classification: identifying whether the subject perceived an erroneous or correct machine output.

To approach this, classical EEG studies typically adopt event-related potential (ERP) analysis, aligning EEG signals to error onset and averaging across trials to reveal a sharp negative deflection approximately 50 to 100 ms after erroneous responses. This component is predominantly distributed over fronto-central regions, with maximal amplitude observed at midline electrodes such as FCz, reflecting rapid neural responses to perceived errors. [27] These indicate that ERN related tasks require modeling fast transient temporal dynamics while preserving spatial sensitivity to specific scalp locations in order to capture localized error related activity.

B. Stress: Psychological Stress Recognition

This longitudinal dataset assesses psychological stress in a naturalistic educational setting. EEG data were recorded from 18 students during a 5-minute resting state prior to attending classes over an entire semester, resulting in 92 distinct recording sessions. Ground truth labels were derived from the Depression, Anxiety, and Stress Scale (DASS) questionnaires completed by the participants immediately before the EEG recordings. In our benchmark, the continuous EEG data are segmented into 5-second epochs to perform a binary classification task, distinguishing between stressed and normal states.

Traditional EEG-based stress studies primarily rely on spectral features. Increased psychological stress has been associated with reduced alpha band power, together with elevated delta and theta power over frontal regions. Concurrently, beta band power increases have been reported in temporal and parietal areas, suggesting enhanced activation of task relevant cortical regions under stress. [28] To jointly characterize temporal evolution and frequency specific changes, classical approaches such as wavelet transforms are commonly applied to EEG signals to extract instantaneous energy variations across frequency bands. [29] These analyses demonstrate the need for joint temporal and spectral modeling across multiple cortical regions, capturing both dynamic time varying EEG activity and power spectrum alterations to characterize physiological patterns associated with stress.

C. Movement: Real World Multiactivity Decoding

To evaluate motor decoding in unconstrained environments, we collected a real world multi-activity dataset. Subjects were instructed to perform six distinct daily activities: walking straight, walking in an S shape, head nodding, head shaking, picking up objects, and climbing stairs (performed outdoors). Each activity was executed at both slow and fast paces. Following noise artifact removal, the continuous recordings are segmented into 1-second epochs. The downstream task is formulated as a 6 class classification to decode the specific type of movement being executed.

In motor-related EEG tasks such as motor imagery, classical analyses focus on modulation of sensorimotor rhythms. During imagined or executed movements, power in the contralateral mu (8 to 13 Hz) and beta (approximately 13 to 30 Hz) bands decreases, a phenomenon known as event-related desynchronization (ERD). Following movement termination, beta band activity typically exhibits rebound enhancement, referred to as event-related synchronization (ERS). These effects are mainly observed over sensorimotor electrodes such as C3 and C4. [30] Consequently, motor EEG tasks require joint time-frequency modeling to capture power changes aligned to movement onset and offset, together with spatial modeling centered on sensorimotor cortical channels to represent rhythmic modulation associated with motor processes.

D. SEED-VII: Continuous Emotion Recognition

The SEED-VII dataset investigates dynamic emotional responses using a multimedia elicitation paradigm. Subjects watched 80 carefully selected video clips designed to evoke seven emotional states (happy, sad, fear, disgust, surprise, anger, and neutral), such as utilizing magic tricks to elicit surprise. Following the experiment, participants continuously annotated their perceived emotional intensity every 4 seconds using a mouse wheel, providing labels ranging from 0 to 1. For our evaluation, we partition the data into 4 second segments. We exclude segments labeled as neutral and those with an emotional intensity below 0.5, formulating the downstream task as a 6 class basic emotion classification.

Emotion-related EEG analyses commonly employ classical measures such as frontal asymmetry and event-related potentials. Hemispheric asymmetry in frontal alpha power reflects approach versus withdrawal tendencies: relatively reduced alpha power over the left frontal cortex, corresponding to increased cortical activity, is typically associated with positive or approach-related emotions, whereas stronger right frontal activity is linked to negative or withdrawal-related emotions. [31] In addition, emotional stimuli reliably evoke a late positive potential (LPP), characterized by enhanced positive ERP amplitudes over parietal and occipital regions approximately 400 to 600 ms after stimulus onset, with magnitude increasing as emotional intensity rises for both positively and negatively valenced items. [32] These conventional analyses indicate that emotion-related EEG requires spatial modeling of hemispheric activity differences and temporal modeling of characteristic ERP components such as the LPP.

E. ISRUC: Sleep Stage Scoring

To evaluate clinical sleep monitoring, we utilize the healthy control subgroup of the ISRUC Sleep dataset. Ten healthy subjects underwent polysomnography recordings while sleeping naturally in a hospital ward, without any artificial sensory stimulation. The continuous EEG recordings were scored by two human experts into five standard sleep stages: Awake (W), N1 (drowsiness/transitional sleep), N2 (light sleep), N3 (deep or slow wave sleep), and Rapid Eye Movement (REM). We segment these recordings into standard 30-second epochs to perform a 5-class sleep stage classification.

N3 is dominated by sustained large scale slow wave activity. N2 is defined by transient events such as sleep spindles and K complexes, which are locally generated but often propagate across multiple cortical regions. N1 is characterized by attenuated alpha activity and low amplitude mixed frequency patterns, while REM exhibits wake like EEG with phasic events and distributed cortical activation. [33] [34] [35] Reliable discrimination requires integrating transient local events, multi channel spatial coherence, and long range temporal context across consecutive epochs.

III. Methodology

A. Data Splitting

Data splitting followed two distinct protocols. For the SEED-VII dataset, we adopted a subject-independent approach to evaluate cross-

TABLE III
HYPERPARAMETER SETTINGS: LEARNING RATE, WEIGHT DECAY, AND BATCH SIZE FOR EACH MODEL ON EACH DOWNSTREAM TASK.

Model	Hyperparameter	KaggleERN	Stress	Movement	Seed-VII	ISRUC
LaBraM	Learning Rate	[0.00009, 0.0005, 0.0009]	[0.0001, 0.0005, 0.001]	[0.00009, 0.0005, 0.0009]	[0.00009, 0.0005, 0.0009]	[0.00009, 0.0005, 0.0009]
	Weight Decay	[0.0001, 0.05]	[0.001, 0.05]	[0.0001, 0.05]	[0.0001, 0.005, 0.05]	[0.0001, 0.005, 0.05]
	Batch Size	[256, 512, 1024]	[256, 512, 1024]	[128, 256, 512]	[128, 256, 512]	[128, 256, 512]
CBraMod	Learning Rate	[0.00005, 0.0001, 0.0005]	[0.00005, 0.0001, 0.0005]	[0.00005, 0.0001, 0.0005]	[0.00005, 0.0001, 0.0005]	[0.00005, 0.0001, 0.0005]
	Weight Decay	[0.00002, 0.002]	[0.00002, 0.002]	[0.00002, 0.002]	[0.00002, 0.002]	[0.00002, 0.002]
	Batch Size	[64, 256, 512]	[64, 256, 512]	[64, 128, 256]	[64, 256, 512]	[64, 256, 512]
BIOT	Learning Rate	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]
	Weight Decay	[0.00001, 0.001]	[0.00001, 0.001]	[0.00001, 0.001]	[0.00001, 0.001]	[0.00001, 0.001]
	Batch Size	[256, 512, 1024]	[256, 512, 1024]	[256, 512, 1024]	[256, 512, 1024]	[256, 512, 1024]
EEGPT	Learning Rate	[0.0001, 0.0004, 0.001]	[0.0001, 0.0004, 0.001]	[0.0001, 0.0004, 0.001]	[0.0001, 0.0004, 0.001]	[0.0001, 0.0004, 0.001]
	Weight Decay	[0.001, 0.01]	[0.001, 0.01]	[0.001, 0.01]	[0.001, 0.01]	[0.001, 0.01]
	Batch Size	[32, 64, 128]	[32, 64, 128]	[32, 64, 128]	[32, 64, 128]	[32, 64, 128]
NeuroGPT	Learning Rate	[0.0003, 0.0001, 0.00006]	[0.0003, 0.0001, 0.00006]	[0.0003, 0.0001, 0.00006]	[0.0003, 0.0001, 0.00006]	[0.0003, 0.0001, 0.00006]
	Weight Decay	[0.01, 0.1]	[0.01, 0.1]	[0.01, 0.1]	[0.01, 0.1]	[0.01, 0.1]
	Batch Size	[16, 32, 64]	[16, 32, 64]	[32, 64, 128]	[32, 64, 128]	[32, 64, 128]
STTrans.	Learning Rate	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]	[0.0005, 0.001, 0.005]
	Weight Decay	[0.00001, 0.001]	[0.00001, 0.001]	[0.00001, 0.001]	[0.00001, 0.001]	[0.00001, 0.001]
	Batch Size	[256,512,1024]	[256,512,1024]	[256,512,1024]	[256,512,1024]	[256,512,1024]
NeuroLM	Learning Rate	[0.000005, 0.00001, 0.00005, 0.0005]	[0.00001, 0.00005, 0.0005]	[0.000005, 0.00001, 0.00005, 0.0005]	[0.000005, 0.00001, 0.00005, 0.0005]	[0.000005, 0.00001, 0.00005, 0.0005]
	Weight Decay	[0.0, 0.01, 0.1]	[0.0, 0.01, 0.1]	[0.0, 0.01, 0.1]	[0.0, 0.01, 0.1]	[0.0, 0.01, 0.1]
	Batch Size	eeg=[52, 48, 36] text=[13, 12, 9]	eeg=[52, 48, 36] text=[13, 12, 9]	eeg=[52, 48, 36] text=[13, 12, 9]	eeg=[52, 48, 36] text=[13, 12, 9]	eeg=[52, 48, 36] text=[13, 12, 9]

subject generalization, assigning subjects 1–18 to training, subject 19 to validation, and subject 20 to testing. For all other datasets, we applied a stratified random split on the pooled samples, partitioning the data into 80% training, 10% validation, and 10% testing sets.

B. Preprocessing Pipeline

We contend that foundation models are inherently coupled with their original preprocessing choices: model representations are learned on data characteristics specific to pretraining preprocessing protocols. Consequently, applying uniform preprocessing across models may introduce distribution shift, contributing to performance discrepancies between original publications and cross-benchmark evaluations. To mitigate this problem, we tailored preprocessing pipelines to each model's pretraining protocol, matching filtering parameters, artifact rejection thresholds, and normalization schemes to minimize distribution mismatch between pretraining and downstream evaluation as shown in Table IV.

TABLE IV
PREPROCESSING SUMMARY

Model	Our Preprocessing
LaBraM	Resample to 200 Hz; bandpass 0.1–75 Hz; notch 50 or 60 Hz; amplitude normalization
CBraMod	Bandpass 0.3–75 Hz; notch 50 or 60 Hz; resample to 200 Hz; amplitude normalization
BIOT	Resample to 200 Hz; amplitude normalization
EEGPT	Resample to 256 Hz; bandpass 0.1–75 Hz; notch 50 or 60 Hz; DC offset removal; average reference; amplitude normalization
NeuroGPT	Resample to 250 Hz; bandpass 0.5–100 Hz; notch 50 or 60 Hz; DC offset removal; linear detrending; average reference; amplitude normalization
STTrans	Resample to 250 Hz; bandpass 4–40 Hz; notch 50 or 60 Hz; amplitude normalization
NeuroLM	Resample to 200 Hz; bandpass 0.1–75 Hz; notch 50 or 60 Hz; amplitude normalization

C. Fine-tuning Strategies

Existing benchmarking efforts often adopt default hyperparameters from original publications or apply uniform settings across all models. These approaches fail to account for model-specific sensitivities and dataset characteristics. To ensure fair comparison, we conducted systematic hyperparameter searches for each model on each task, optimizing learning rate, batch size, weight decay, and task-specific architectural parameters through grid search as shown in Table III.

D. Model Selection and Evaluation Metrics

To ensure robust and fair evaluation across diverse datasets, particularly those exhibiting severe class imbalance, we standardized our training objectives and evaluation metrics. Binary classification tasks were optimized using binary cross-entropy loss equipped with class reweighting, while multi-class tasks employed standard cross-entropy loss adjusted for class distributions.

SEED-VII presented unique challenges due to the sparse temporal distribution of emotionally salient responses within long recordings. We developed a tailored protocol leveraging the dataset's continuous intensity annotations (updated every 4 seconds) to extract 4-second EEG segments with intensity greater than 0.5, ensuring training focuses on emotionally relevant periods. The neutral class is excluded in the evaluation, and a video-level evaluation through majority voting was implemented: each test video was decomposed into 4-second chunks, and chunk predictions were aggregated via majority vote. Tied predictions received partial credit (0.5) if the ground-truth label is among tied classes, otherwise 0. For video evaluation, we first aggregate the predictions of 4-second segments within the same video via majority voting to obtain the overall video prediction. We then calculate the balanced accuracy (the average of recall across all classes) using these video predictions and the ground truth labels. In the event of a tie during voting, the first label among the tied candidates is adopted as the final prediction for metric computation.

During our hyperparameter search, we selected model checkpoints based on their peak performance on the validation set. The final reported results reflect the selected model's performance on the held-out test set. Across all benchmark tasks, we uniformly report balanced accuracy, which is calculated as the mean per-class recall.

E. Statistical Analysis

To rigorously evaluate whether the observed performance differences between models reflect true architectural superiority rather than

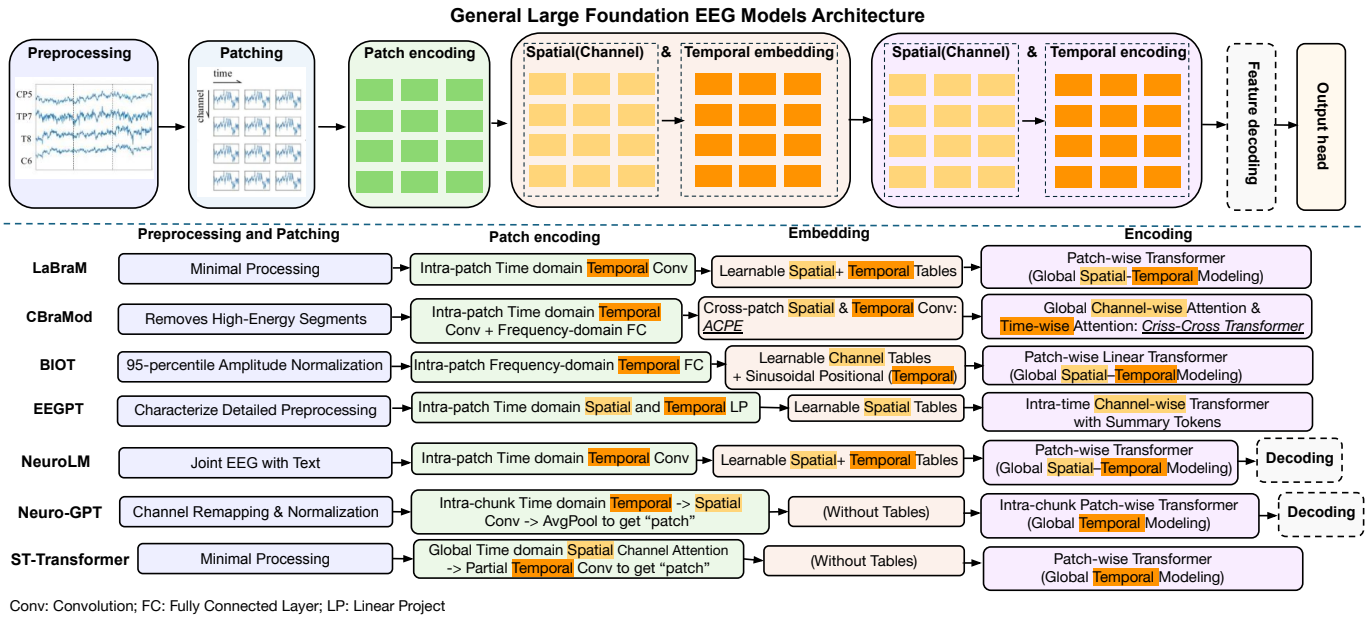

Fig. 4. Architectural comparison of large EEG foundation models. Top: a unified pipeline shared by most models, including preprocessing, patch construction, patch encoding, spatial (channel) and temporal embedding and encoding, decoding, and task heads. Bottom: architecture for LaBraM, CBraMod, BIOT, EEGPT, NeuroLM, NeuroGPT, and ST-Transformer, detailing their feature extraction, positional encoding strategies, and Transformer attention scope. The figure contrasts local versus cross-patch modeling, temporal versus spatial (channel-wise) feature extraction, and different attention formulations.

random variance from hyperparameter optimization, we conducted sample-level statistical significance testing.

We used McNemar's test for paired nominal data, which is designed for comparing two classifiers evaluated on the same test set. For a given pair of evaluated models, we recorded their sample-level prediction outcomes (correct or incorrect). Let b denote the number of samples where the first model predicted correctly while the second model failed, and c denote the number of samples where the first model predicted incorrectly while the second model succeeded.

Under the null hypothesis that the two classifiers have equivalent error rates (i.e., $b = c$), we computed the McNemar statistic with a continuity correction as follows:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

The two-sided p-value was then obtained as the upper-tail probability $P(\chi^2(1) \geq \chi^2)$ utilizing the chi-squared survival function with one degree of freedom. This standardized testing protocol was systematically applied to key pairwise model comparisons across our benchmark tasks to ensure the reliability of our empirical conclusions.

IV. Model Architectures and Pretraining

TABLE VI
Backbone Parameter Scale of Evaluated Models on SEED VII Task

Model	Backbone Parameters
LaBraM	5.8M
CBraMod	4.9M
BIOT	3.2M
EEGPT	25.3M
NeuroGPT	0.16M
NeuroLM	210.2M
ST Transformer	3.5M

We reviewed the architectures' pretraining strategies of seven Large EEG Models, focusing on their Encoder, Quantizer, Decoder, self-supervised objectives, architectural choices, and motivations grounded in EEG data characteristics. The high-level architectural similarities and model-specific components are systematically compared in Figure 4 , while the diverse self-supervised pretraining objectives and the semantic meaning of their respective loss functions are detailed in Table V.

Additionally, Table VI summarizes the overall backbone parameter scale for each evaluated model on SEED VII task. A comprehensive parameter breakdown for the individual components of each architecture is provided in Appendix A.

1) LaBraM

1) Architectures: LaBraM processes each patch through three 1D convolutional layers with group normalization and GELU activation to extract local temporal features within the patch. To encode positional information, the model adds two learnable embeddings to each patch: a temporal embedding based on the patch's time index and a spatial embedding based on channel identity. These position-enriched patch embeddings are then processed by a standard Transformer encoder, which applies self-attention across all patches to model global spatiotemporal dependencies. During downstream fine-tuning, only the convolutional encoder, positional embeddings, and Transformer encoder are retained; pretraining components (vector-quantized tokenizer and spectral decoder) are discarded.
2) Pretraining Task: LaBraM uses a two-stage self-supervised pretraining strategy built around masked token prediction. It operates on a learned EEG vocabulary, where the model learns to reconstruct withheld EEG content by predicting discrete token codes. First, a vector-quantized neural tokenizer is trained to create a discrete EEG vocabulary. This tokenizer uses vector-quantized temporal-spectrum prediction: an autoencoder compresses each EEG patch into a code (from a learned codebook) by learning to reconstruct both the amplitude and phase of

TABLE V
SUMMARY OF SELF-SUPERVISED PRETRAINING OBJECTIVES AND LOSS SEMANTICS ACROSS LARGE EEG FOUNDATION MODELS.

Model	Self supervised method	Pretraining task type	Meaning of losses
LaBraM	Vector quantized spectrum reconstruction then masked neural token prediction	Mask reconstruction	Force a discrete neural vocabulary and make the encoder learn conditional structure of the signal such that latent features capture generative factors that allow reconstructing masked content at the neural token level
CBraMod	Masked raw patch reconstruction from visible context	Mask reconstruction	Make the encoder learn conditional structure of the raw waveform itself so that latent features capture continuous morphology needed to reconstruct masked raw patches
BIOT	Masked based reconstruction **in representation space**	Mask representation matching	Encourage the representation of the perturbed masked input to be close to the representation of the clean full input
EEGPT	Masked based reconstruction **in representation space** and time domain	Mask representation matching; Mask reconstruction	Encourage the representation of the perturbed masked input to be close to the representation of the clean full input and make the encoder learn conditional structure of the raw waveform itself so that latent features capture continuous morphology needed to reconstruct masked raw patches
NeuroGPT	Masked segment prediction in embedding space	Mask prediction	Make the encoder learn conditional temporal structure that allow reconstructing the missing embedding
NeuroLM	Vector quantized temporal and frequency reconstruction then adversarial domain alignment then LLM next neural token prediction	Mask prediction	Shape the neural codebook with time and frequency reconstruction align EEG embedding distribution into text embedding space via adversarial alignment and enable an LLM to perform autoregressive modeling over EEG tokens to learn cross channel causal structure

the patch's Fourier spectrum. The resulting codebook (8192 entries of 64 dimensions) provides "neural tokens" that capture salient EEG features. Next, the model is pretrained via masked EEG modeling: it treats the sequence of discrete EEG tokens as input, randomly masks 50% of them, and learns to predict the masked token from the remaining context. [1]

2) CBraMod

1) Architectures: CBraMod employs dual-pathway patch encoding: a time-domain pathway applies 1D convolutions (with normalization and GELU) to extract local temporal features, while a frequency-domain pathway computes the Fast Fourier Transform (FFT) magnitude spectrum and projects it through a fully connected layer. The model then applies Asymmetric Conditional Positional Encoding (ACPE), implemented as a 2D convolution with a (19, 7) kernel over channels and time, which generates dynamic positional embeddings across patches). These ACPE-derived embeddings are added to the dual-pathway features before Transformer processing.
   CBraMod's backbone employs a criss-cross Transformer architecture that factorizes attention into separate spatial and temporal streams. Within each layer, attention heads and feature dimensions are partitioned between two pathways: spatial attention heads attend across channels at each time step to model inter-channel dependencies, while temporal attention heads attend across time steps within each channel to capture long-range temporal dynamics. This factorized design enables explicit modeling of spatial and temporal structure, aligning well with the distinct spatial and temporal characteristics of EEG signals. [16]
2) Pretraining Task: CBraMod is pretrained via a masked EEG reconstruction strategy that directly learns to recover missing raw signals from context. In pretraining, CBraMod randomly masks 50% of the patches (replacing them with a mask token) and feeds the patches into the model. The model's output is passed through a reconstruction head that tries to reconstruct the original time-series of each masked patch. The reconstruction head is a lightweight decoder (a fully-connected layer) that outputs the patch's time domain signal. [16]

3) BIOT

1) Architectures: BIOT constructs patch embeddings through three components. First, each patch undergoes Short-Time Fourier Transform (STFT) to obtain its magnitude spectrum, which is projected through a fully connected layer to a 256-dimensional segment embedding. Second, a learned channel embedding encodes the patch's electrode origin. Third, sinusoidal positional encodings (rather than learned embeddings) mark temporal position within the channel sequence. These three components are summed to form the final patch representation. All patches across channels are concatenated into a single sequence and processed by a Transformer encoder employing linear-complexity attention (Linformer) to handle long sequences efficiently. [24]
2) Pretraining Task: Rather than masking or reconstructing signals, BIOT's pretraining is a contrastive self-prediction approach: the model learns to predict an EEG sample's representation from itself (under distortions), yielding a universal EEG encoder robust to segment token removals and cross-dataset variations. In concrete terms, for each unlabeled EEG sample, one generates a corrupted view $\tilde{S}$ by randomly removing a subset of segment tokens from each channel's sequence. Specifically, BIOT randomly samples segment tokens from each channel without replacement: for each channel, a random number of tokens is selected, creating the perturbed sample. The intact sample S and its corrupted version $\tilde{S}$ are both fed through the BIOT encoder to produce embeddings $Z = f_\theta(S)$ and $\tilde{Z} = f_\theta(\tilde{S})$, respectively. The corrupted embedding $\tilde{Z}$ is then passed through a small predictor MLP (two linear layers with GELU activation) to produce the prediction. BIOT then aligns the predicted embedding from the corrupted view to match the embedding of the original sample using a contrastive loss. [24]

4) EEGPT

1) Architectures: EEGPT segments EEG into patches (250 ms at 256 Hz per channel). Each patch is independently embedded via an intra-patch projection, and the resulting tokens are combined with channel encodings. For each temporal patch,

the model appends $K = 4$ learnable summary tokens (512-dimensional) to the channel token sequence. A Transformer encoder processes this augmented sequence, enabling summary tokens to aggregate cross-channel information through self-attention. The architecture retains only these summary tokens as the final representation, compactly encoding multi-channel activity at each time step. [7]

2) Pretraining Task: EEGPT is pretrained using a dual self-supervised learning framework that combines spatio-temporal representation alignment and mask-based reconstruction.
(1) Mask-based reconstruction: The input EEG is first segmented into patches and a random subset of patches is masked out. The model's encoder processes the partially masked signal (only seeing patches from masked time segments), the predictor predicts summary tokens for all time segments, and then the reconstructor attempts to reconstruct the original time-series waveform for each masked patch location.
(2) Latent representation alignment: In addition to reconstructing the signal, EEGPT explicitly forces the model's internal representations to be predictive of uncorrupted EEG features. Specifically, for all time segments, EEGPT aligns the predicted summary tokens from the predictor (which operates on masked encoder outputs) with the true summary tokens obtained from a momentum encoder pass on the full, unmasked EEG. The alignment loss compares predictor outputs for all time segments (both masked and unmasked) with momentum encoder outputs for all time segments. [7]

5) NeuroGPT

1) Architectures: NeuroGPT uses the encoder part for downstream tasks. The encoder segments input EEG into fixed-length chunks and applies two sequential convolutional layers: a (1 × 25) temporal convolution that extracts local patterns within each channel, followed by a (C × 1) spatial convolution integrating information across all C channels at each time step. The convolutional features undergo batch normalization, ELU activation, temporal average pooling, dropout, and linear projection, and are then processed by Transformer self-attention layers to model intra-chunk dependencies. [25]
2) Pretraining Task: NeuroGPT consists of an EEG encoder coupled with a GPT-2 (decoder-only Transformer) autoregressive decoder. For each original sequence of length N, it generates $N - 1$ training samples, where each sample masks a different position from position 2 to position N. NeuroGPT applies a causal masking scheme where masked chunk embeddings are replaced with a learnable mask token, and an attention mask ensures that the GPT decoder can only attend to chunks preceding the masked position to preserve the causal temporal relationships in EEG signals. The GPT decoder then processes these partially observed sequences and is tasked to generate the masked chunk embedding in each case. [25]

6) NeuroLM

1) Architectures: NeuroLM employs a two-stage architecture comprising a vector-quantized (VQ) tokenizer and an autoregressive language model decoder. The VQ tokenizer applies three convolutional layers to extract temporal features from raw EEG, producing 768-dimensional patch embeddings that are quantized into discrete codes via an 8192-entry codebook. The tokenizer is trained through self-supervised reconstruction of the input signal's time-frequency representation. Critically, the training objective includes adversarial alignment that maps EEG code embeddings into the same vector space as text embeddings, enabling cross-modal integration with language. For downstream tasks, the frozen VQ encoder converts EEG inputs into discrete token sequences, which are processed by a pretrained GPT-2 decoder with expanded vocabulary to include the EEG codebook entries. This architecture enables NeuroLM to leverage pretrained language model capabilities for EEG sequence modeling [9]
2) Pretraining Task: NeuroLM treats EEG as a "foreign language" to leverage large language-model architectures. The pretraining is executed in two stages: Stage 1: Text-aligned neural tokenizer. NeuroLM first learns a discrete token encoding for EEG via vector-quantized temporal-frequency prediction, augmented with adversarial alignment. The model encodes EEG into patch tokens and trains decoders to reconstruct each patch in both time-domain and frequency-domain. Crucially, NeuroLM performs EEG–text embedding alignment at this stage: it introduces a domain classifier that tries to distinguish EEG-token embeddings from text embeddings, and trains the tokenizer to make EEG token embeddings indistinguishable from text embeddings to the domain classifier via a gradient reversal layer. By feeding in sample text token embeddings from a frozen GPT-2 word token embedding layer during tokenizer training, NeuroLM forces the EEG codes to occupy a feature space aligned with natural language tokens, bridging EEG data with the semantic space of an LLM without needing any explicit EEG-text pairs. Stage 2: Multi-channel autoregressive pretraining. After learning the VQ tokenizer, NeuroLM freezes the tokenizer's encoder and uses it to convert all EEG data into sequences of discrete tokens. It then trains a causal Transformer (initialized with GPT-2 weights) to model these token sequences in an autoregressive manner, performing next-token prediction on EEG token sequences. The Transformer is then trained on the objective $p_\theta(z_{t+1}|z_{\leq t})$: predict the next EEG code token given all previous tokens across all channels. [9]

7) ST-Transformer

1) Architectures: ST-Transformer employs a two-stage spatial-then-temporal architecture without explicit patch tokenization. First, a channel attention mechanism computes importance weights across all EEG channels: query and key representations are generated via linear transformations, and attention weights aggregate cross-channel information into spatially-pooled representations. The spatially-attended signal is then processed through convolutional layers (producing 256-dimensional features) that segment the signal into temporal patches. A Transformer encoder applies self-attention across these temporal patches to model long-range dependencies. [1] [26]
2) Pretraining Task: The ST-Transformer (Spatial-Temporal Transformer) represents an earlier transformer-based EEG model that did not rely on large-scale self-supervised pretraining.

V. Model Pretraining Datasets and Preprocessing

1) Preprocessing Pipeline for Models

Here, we summarize the data preprocessing strategies employed by the seven models during their pretraining phases in table VII

Models such as LaBraM and NeuroLM intentionally adopt minimal preprocessing, reflecting a design philosophy that prioritizes preservation of the raw temporal–spatial structure to promote robust cross-dataset generalization. Their pipelines typically involve resampling (e.g., to 200 Hz), a 0.1–75 Hz bandpass filter, notch filtering at local

[1] We use ST transformer implementation from BIOT [24], which omits CSP spatial filtering for computational efficiency.

TABLE VII
DETAILED PREPROCESSING PIPELINES AND INPUT CONFIGURATIONS FOR THE SEVEN EEG FOUNDATION MODELS

Model	Filtering & Resampling	Artifacts, Channel & Reference	Normalization & Input
LaBraM	200 Hz BP: 0.1–75 Hz Notch: Yes	–	Norm: Scale to $[-1, 1]$ Win: 4 or 8 sec Patch: $1s \times 1ch$
CBraMod	200 Hz BP: 0.3–75 Hz Notch: Yes	Bad Time Removal Ch: 19 (10-20 system)	Norm: Scale to $[-1, 1]$ Win: 30 sec Patch: $1s \times 1ch$
BIOT	200 Hz (No BP/Notch specified)	–	Norm: 95th-percentile abs Win: 10 sec Patch: $1s \times 1ch$
EEGPT	256 Hz BP: Dataset specific Notch: Yes	DC offset Ref: Global average Ch: 58 unified	Norm: EA (MI) / Z-score (Sleep) Win: 4 or 10 sec Patch: $250ms \times 1ch$
NeuroGPT	250 Hz BP: 0.5–100 Hz Notch: Yes	Bad Ch Interpolation DC offset Trend Removal Ref: Average Ch: 22 (10-20 system)	Norm: Per channel Z-score Win: 2 sec Patch: $2s \times 22ch$
NeuroLM	200 Hz BP: 0.1–75 Hz Notch: Yes	–	Norm: Scale to $[-1, 1]$ Win: L (Variable) Patch: $1s \times 1ch$
ST-Transformer	250 Hz BP: 4–40 Hz (No Notch)	–	Norm: Global Z-score Win: 4 sec Patch: $10pts \times 1ch$

Note: **BP**: bandpass filter; **Ref**: Re-referencing; **Ch**: Channel;
Interp.: Interpolation; **Norm**: Amplitude Normalization; **Win**: Time Window;
EA: Euclidean Alignment; **MI**: Motor Imagery task;

line-noise frequencies (50 Hz or 60 Hz), and amplitude normalization that constrains signals to the range -1 to 1. [1] [9]

ST-Transformer follows a similarly lightweight approach, relying on 250 Hz resampling, a 4–40 Hz bandpass filter, z-score standardization, and removal of EOG channels. [26]

CBraMod's data are resampled to 200 Hz, bandpass filtered from 0.3–75 Hz, notch-filtered, and select 19 channels from 10-20 system (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2) used during pretraining. EEG amplitudes are normalized by setting the unit scale to 100 μV, ensuring values fall between -1 and 1. A key step in CBraMod is the exclusion of frames whose absolute amplitudes exceed 100 μV, treating them as bad-time intervals. This lightweight cleaning strategy yields a relatively clean corpus for robust pretraining. [16]

The BIOT pipeline resamples all signals to 200 Hz and applies a 95th-percentile amplitude normalization to reduce cross-dataset scaling discrepancies. [24]

In contrast, EEGPT and NeuroGPT adopt the most elaborate preprocessing strategies among recent large EEG models. In EEGPT, the authors apply dataset-specific pipelines tailored to the characteristics and artifact profiles of each corpus. These include multi-stage bandpass filtering (0–38 Hz for MI downstream tasks, 30 Hz low-pass filter for Sleep-EDFx, and 120 Hz low-pass filter for PhysioNetP300), DC offset correction, global average referencing, channel unification, and per-channel normalization (e.g., EA normalization or z-score normalization). [7]

NeuroGPT first selects 22 channels derived from the international 10–20 system (Fp1, Fp2, F7, F3, Fz, F4, F8, T1, T3, C3, Cz, C4, T4, T2, T5, P3, Pz, P4, T6, O1, Oz, O2). Channels with zeroed or missing signals are reconstructed via spatial interpolation using averages from neighboring electrodes. Subsequent steps include average rereferencing, resampling to 250 Hz, 0.5–100 Hz bandpass filtering, notch filtering, DC offset correction, linear trend removal, and z-transform normalization. For the downstream dataset, data are transformed into the pretraining configuration using a transformation matrix. [25]

2) Pretraining Dataset

Foundation models were pretrained on multiple datasets spanning **BCI**, **cognitive** and **clinical** applications (Table VIII). Each dataset is characterized by its sampling rate (Hz), number of channels, subject count, segment duration (length of individual epochs), task type, total recording length, and which models used it for pretraining.

BCI datasets primarily cover three paradigms: motor imagery, P300, and SSVEP. Due to the controlled laboratory settings required for BCI experiments, these datasets are limited in scale compared to clinical data, most containing fewer than 50 subjects and under 50h of recordings.

Cognitive datasets are dominated by emotion recognition tasks, with additional tasks including visual categorization and resting states. These datasets employ consistent high-density montages (62-64 channels) typical of research-grade configurations. Although some datasets appear large (e.g., SEED Series with 170.5h, Self-collected Corpus with 342.2h), these are aggregations of multiple smaller experiments across different emotion categories or task types. The individual cognitive experiments remain small in scale due to the need for strictly controlled stimuli and experimental conditions.

Compared to the other two applications, most clinical datasets benefit from large-scale hospital data collection pipelines, achieving massive scale (up to 14,987 subjects and 24,000h).

LaBraM and NeuroLM were pretrained on the broadest range of data, incorporating clinical, BCI, and cognitive datasets to capture diverse EEG signal characteristics across applications. [1] [9] The original BIOT study released multiple sets of pretrained weights, including both supervised and unsupervised versions. Because all other foundation models evaluated in our benchmark rely exclusively on self-supervised pretraining, utilizing BIOT's supervised weights would introduce an unfair advantage. Therefore, to ensure a rigorous and fair comparison, we specifically selected the BIOT weights pretrained on the PREST dataset using solely their unsupervised contrastive learning objective. CBraMod, BIOT, and NeuroGPT focused primarily on clinical datasets, specializing in tasks such as seizure detection, sleep staging, and abnormal event classification. [16] [24] [25] EEGPT was pretrained exclusively on BCI and cognitive datasets, emphasizing motor imagery, P300 paradigms, and emotion recognition tasks. [7] STTransformer differs from other models as it does not include a pretraining stage and is trained directly on downstream tasks. [26]

VI. RESULTS

Comparing our custom protocol on SEED-VII and multi-class classification on video segments presents improvements as shown in Table X.

TABLE X
COMPARING PERFORMANCE(% BACC) OF DIFFERENT STRATEGY ON SEED-VII DATASETS.

Dataset	Strategy	Model			
		LaBraM	**CBraMod**	**EEGPT**	**NeuroGPT**
SEED-VII	Voting + remove neutral + continuous annotations (4 s)	30.83	27.52	41.29	27.68
	Original (no voting, all data included, video-level annotations)	21.03	24.52	19.62	15.18

Table IX presents performance across five tasks under linear probing and full fine-tuning with optimized hyperparameters. Fine-tuning yields substantial improvements across all models, with gains

TABLE VIII
OVERVIEW OF PRETRAINING DATASETS

Model	Dataset Name	Application	Category	Rate (Hz)	Channels	Subjects	Duration	Task (Classes)	Length (h)
NeuroLM	TUEG (TUH Corpus) [36]	Clinical	Abnormal/Seizure	250-1024	17-23	14,987	30 s	N/A	∼24,000
	SEED Series [37]–[39]	Cognitive	Emotion Recog.	1000	62	51	4-10 s	4 (SEED-IV), 5 (SEED-V), 3 (SEED-GER), 3 (SEED-FRA)	170.5
	BCI Comp. IV-1 [40]	BCI	Motor Imagery	1000	59	7	-	2 (L/R Hand, Foot)	8.2
	Emobrain [41]	Cognitive	Emotion Recog.	1024	64	16	-	3 (Emotions)	4.9
	Grasp and Lift [42]	BCI	Motor Control	500	32	12	-	6 (motions)	11.7
	Inria BCI [19]	BCI	P300/Spelling	600	56	26	1.3 s	2 (Error vs Correct)	30.0
	PhysioNet-MI [43]	BCI	Motor Imagery	160	64	109	4 s	4 or 5	47.3
	Raw EEG (Trujillo) [44]	Cognitive	Visual Categori.	256	64	48	-	2	34.4
	Resting (Trujillo) [45]	Cognitive	Resting State	256	64	22	480 s	2 (Eyes Open/Closed)	3.0
	Siena Scalp [46]	Clinical	Seizure/Resting	512	31	14	-	2 (Seizure Detection)	30.5
	SPIS Resting [47]	Cognitive	Attention/Resting	2048	64	10	-	2 (Eyes Open/Closed)	0.8
	Target vs Non-Target [48]	BCI	P300	512	32	50	-	2 (Target/Non)	16.0
	Self-collected Corpus [49]–[53]	Cognitive	Mixed (4 Emotion Recog. + 1 Confidence Classi.)	1000	62	>140	-	7, 2, 3, 4 (Emotions); 5 (Confidence Levels)	342.2
NeuroGPT	TUEG (TUH Corpus) [36]	Clinical	Abnormal/Seizure	250-1024	17-23	14,987	30 s	N/A	∼24,000
CBraMod	TUEG (TUH Corpus) [36]	Clinical	Abnormal/Seizure	200 (resampled)	19	14,987	30 s	N/A	>9,000
LaBraM	TUSZ [54]	Clinical	Seizure Detection	256	19-23	675	-	Seizure Events	1,138.5
	TUEP [55]	Clinical	Epilepsy Diagnosis	256	19-23	200	-	2 (Epilepsy/No)	591.2
	TUAR [56]	Clinical	Artifact Detection	256	23	213	-	5 (Artifacts)	92.2
	TUSL [57]	Clinical	Slowing Event	256	23	38	10 s	3 (Slowing types)	20.6
	Siena Scalp [46]	Clinical	Seizure/Resting	512	31	14	-	2 (Seizure Detection)	30.5
	SEED Series [37], [39], [58]	Cognitive	Emotion Recog.	1000	62	46	4-10 s	3 (SEED), 4 (SEED-IV), 3 (SEED-GER), 3 (SEED-FRA)	166.75
	BCI Comp. IV-1 [40]	BCI	Motor Imagery	1000	59	7	-	2 (L/R Hand, Foot)	8.2
	Emobrain [41]	Cognitive	Emotion Recog.	1024	64	16	-	3 (Emotions)	4.9
	Grasp and Lift [42]	BCI	Motor Control	500	32	12	-	6 (motions)	11.7
	Inria BCI [19]	BCI	P300/Spelling	600	56	26	1.3 s	2 (Error vs Correct)	30.0
	PhysioNet-MI [43]	BCI	Motor Imagery	160	64	109	4 s	4 or 5	47.3
	Raw EEG (Trujillo) [44]	Cognitive	Visual Categori.	256	64	48	-	2	34.4
	Resting (Trujillo) [45]	Cognitive	Resting State	256	64	22	480 s	2 (Eyes Open/Closed)	3.0
	SPIS Resting [47]	Cognitive	Attention/Resting	2048	64	10	-	2 (Eyes Open/Closed)	0.8
	Target vs Non-Target [48]	BCI	P300	512	32	50	-	2 (Target/Non)	16.0
	Self-collected Corpus [49]–[53]	Cognitive	Mixed (4 Emotion Recog. + 1 Confidence Classi.)	1000	62	>140	-	7, 2, 3, 4 (Emotions); 5 (Confidence Levels)	342.2
BIOT	PREST [24]	Clinical	Seizure/Abnormal	200	16	6,478	10 s	N/A	14,197
EEGPT	PhysioNet-MI [43]	BCI	Motor Imagery	160	64	109	4 s	4 or 5	47.3
	HGD [59]	BCI	Motor Imagery	256	128	14	4 s	4 (L/R Hand, Feet, Rest)	15.6
	TSU [60]	BCI	SSVEP	256	64	35	4 s	40 (Characters)	9.3
	SEED [58]	Cognitive	Emotion Recog.	256	62	15	10 s	3 (Emotions)	∼45
	M3CV [61]	BCI	Multi-paradigm	256	∼64	106	4 s	14 (Task Types)	167.5

varying by model and task (e.g., CBraMod: 26.75% → 78.4% on ISRUC; more robust baselines: 5-15% average gain). Critically, fine-tuning alters model rankings: several models that underperform in linear probing achieve competitive or superior performance after optimization. We further observe clear task-dependent performance patterns: LaBraM and CBraMod excel on KaggleERN, Stress, and Movement classification tasks, while EEGPT and BIOT achieve top performance on SEED-VII and ISRUC. We analyze these architectural and task-specific effects in Section VI.

While our benchmark highlights the strengths of different architectures, the pursuit of a truly universal foundation model faces ongoing hurdles. A more detailed breakdown of these research gaps can be found in the Outstanding Questions section.

VII. DISCUSSION

1) Impacts of Architecture

To characterize the relationship between architecture and performance, we first need to understand what physiological features these tasks require the analysis method to extract. In other words, we need to identify which traditional non-machine learning methods are commonly used for these five tasks as shown in Table II. By jointly examining Table II, Figure 4, and the performance results in Table IX, several consistent patterns emerge across models and tasks.

Among the seven evaluated models, the four more recent architectures (LaBraM, CBraMod, BIOT, and EEGPT) achieve top-tier performance (purple regions) on at least two of five tasks, while earlier models (NeuroGPT and ST-Transformer) and the LLM-based NeuroLM show comparatively weaker results. Notably, model rankings exhibit clear task-dependent patterns that align with architectural design choices.

LaBraM and CBraMod achieve the best fine-tuned performance on three tasks (KaggleERN, Stress, and Movement), which share a common characteristic: spatially localized signals. KaggleERN targets error-related negativity concentrated in fronto-central electrodes (e.g., FCz) within narrow post-error windows; movement classification discriminates signals from contralateral sensorimotor channels (C3/C4); and stress detection relies on band-power changes over limited frontal, temporal, and parietal regions (Table II). In contrast, SEED-VII emotion recognition and ISRUC sleep staging involve broader spatial integration across distributed brain regions with longer temporal context. On these two tasks, EEGPT and BIOT achieve top performance, surpassing LaBraM and CBraMod.

These performance patterns align with architectural differences in patch encoding. LaBraM and CBraMod employ convolutional operations for local feature extraction: LaBraM applies temporal convolutions within patches, while CBraMod uses 2D convolutions (kernel size 19×7) in its ACPE module to explicitly extract local spatiotemporal patterns before Transformer encoding. In contrast, EEGPT employs linear projection, and BIOT uses STFT followed by fully connected layers—both introducing minimal convolutional inductive bias. Our results suggest that tasks with localized spatiotemporal patterns benefit from convolutional feature extraction, while tasks requiring integration across distributed regions benefit from preserving more complete input information for Transformer-based global modeling over longer segments (e.g., 3 seconds).

Within the localized-pattern tasks, CBraMod consistently outperforms LaBraM on event-related paradigms (KaggleERN and Movement). This advantage likely stems from CBraMod's criss-cross Transformer architecture, which factorizes attention into separate temporal (within-channel) and spatial (across-channel) streams rather

TABLE IX
BENCHMARKING RESULTS ACROSS FIVE DOWNSTREAM TASKS (% BACC). LINEAR PROBING RESULTS (FIRST ROW PER TASK) SHOW PERFORMANCE WITH DEFAULT CONFIGURATIONS; BLUE SHADING INDICATES TOP MODELS (DARKER = BEST). FINE-TUNING RESULTS (SECOND ROW) SHOW OPTIMIZED PERFORMANCE; PURPLE SHADING INDICATES TOP MODELS. BEST PERFORMANCE AND MEAN ± STD ACROSS RUNS ARE REPORTED.

Dataset	Metric	Model						
		LaBraM	CBraMod	BIOT	EEGPT	NeuroGPT	STTrans.	NeuroLM
KaggleERN	Linear Probe	**64.99**	54.37	57.57	52.89	54.89	-	-
	FT Best	70.26	**72.02**	60.38	56.64	61.24	67.50	69.03
	FT Mean ± SD	67.7 ± 2.25	62.78 ± 9.37	55.0 ± 2.36	53.8 ± 1.47	59.88 ± 1.19	59.88 ± 3.7	55.38 ± 5.45
Stress	Linear Probe	50.0	50.55	**76.26**	62.65	39.58	-	-
	FT Best	**97.0**	91.35	90.78	69.98	73.65	62.51	62.08
	FT Mean ± SD	85.0 ± 4.67	75.31 ± 11.07	66.83 ± 12.28	59.23 ± 6.16	70.03 ± 2.2	52.53 ± 6.64	50.54 ± 3.88
Movement	Linear Probe	**92.97**	68.36	60.88	64.92	40.69	-	-
	FT Best	98.39	**99.05**	81.01	74.32	86.64	78.02	91.54
	FT Mean ± SD	97.81 ± 0.59	98.76 ± 0.21	69.17 ± 14.3	68.64 ± 3.62	81.72 ± 3.33	74.43 ± 2.72	79.55 ± 10.63
Seed-VII	Linear Probe	23.31	20.39	**24.39**	12.08	14.22	-	-
	FT Best	30.83	27.52	40.18	**41.29**	27.68	27.99	35.11
	FT Mean ± SD	27.80 ± 1.53	21.31 ± 3.84	25.10 ± 6.86	26.60 ± 8.01	24.79 ± 2.61	22.97± 3.52	23.94 ± 5.78
ISRUC	Linear Probe	53.08	26.75	80.86	**82.9**	56.41	-	-
	FT Best	75.80	78.40	82.16	**83.35**	80.30	76.92	76.67
	FT Mean ± SD	76.55 ± 0.92	70.30 ± 5.99	80.15 ± 1.69	81.89 ± 3.6	77.98 ± 1.06	58.66±10.29	67.46 ± 5.89

FT: Full Fine-tuning; SD: Standard Deviation. Full Fine-tuning results present the best performance achieved across the Author Configuration and Hyperparameter Search trials (18 trials for most models; 27 for NeuroLM).

than treating all channel-time tokens uniformly. This factorized design aligns with standard event-related EEG analysis paradigms that examine how individual channels evolve over time, potentially explaining CBraMod's superior performance on ERN and movement decoding.

Linear probing results (blue regions in Table Table IX) reveal a distinct advantage for BIOT, which achieves strong performance on four of five tasks by freezing the pretrained backbone and training only the final linear head. This pattern likely reflects BIOT's frequency-domain patch encoding via STFT, which directly captures spectral features. On sleep staging—a task heavily reliant on frequency-band characteristics—BIOT achieves 80.86% balanced accuracy with linear probing alone, exceeding the full fine-tuned performance of NeuroGPT (80.30%), Cbramod (78.40%), and ST-Transformer (76.92%). Similarly, on stress detection, BIOT's linear probe reaches 76.26%, substantially outperforming other models' linear probing results (next-best: 62.65 %), consistent with stress detection's strong dependence on frequency-band power modulations.

2) Impacts of Pretraining Methods

The third row of each task in Table IX reports mean performance and standard deviation across at least 18 experimental configurations. From these statistics, LaBraM shows relatively stable performance across multiple tasks. In comparison, CBraMod, despite achieving similarly high peak results on KaggleERN, Stress, and Movement, exhibits much larger variance in 4 out of 5 tasks, indicating strong sensitivity to hyperparameter choices.

This stability may stem from LaBraM's VQNSP-based neural tokenizer pretraining, where a codebook of 8192 neural patterns is learned and downstream representations are encouraged to collapse toward this discrete vocabulary. This can enhance expressiveness and provide more robust representations, which may also explain why LaBraM achieves strong linear-probe performance on 3 out of 5 tasks.

NeuroLM exhibits rapid convergence, reaching its peak performance within approximately the first five fine-tuning epochs, substantially faster than other models. Although its final accuracy does not always reach the absolute best, it ranks within the top three on three tasks. This behavior highlights the potential of integrating LLM-based autoregressive modeling with EEG representations. NeuroLM is also the largest model in this benchmark, and the available EEG data volume may be insufficient to fully exploit its capacity. With larger or more diverse pretraining corpora, further performance gains may be achievable.

3) Impacts of Pretraining Data

For error-related potential detection on KaggleERN (Table IX), LaBraM achieves the strongest mean performance, as its pretraining explicitly includes the Inria BCI Challenge dataset, which is identical to KaggleERN. CBraMod, although not pretrained on ERN-related tasks, attains a 1.76% higher best balanced accuracy than LaBraM after hyperparameter search. Similarly, CBraMod achieves performance comparable to or slightly exceeding LaBraM on Stress and Movement ($p = 0.002$). We hypothesize that incorporating more diverse task types during CBraMod pretraining could improve its stability and generalization.

For stress detection, LaBraM achieves both the highest mean and best performance, attributable to its pretraining on the confidence-level dataset, which captures cognitive-affective states associated with mental workload and arousal.

For movement classification, LaBraM attains the best linear-probe performance and achieves results within 0.66% of the top model under full fine-tuning. This advantage is likely directly linked to its pretraining on the motor imagery dataset, enabling robust modeling of motor cortex activation across multiple body regions.

For emotion recognition on SEED-VII, EEGPT achieves the strongest performance because its pretraining corpus includes the SEED series with identical experimental paradigms and electrode configurations. LaBraM, which also uses SEED during pretraining, attains the best mean performance after hyperparameter search. NeuroLM ranks third, benefiting from the same pretraining source.

BIOT demonstrates strong generalization despite lacking emotion-specific pretraining, achieving the best linear-probe performance and reaching within 1.11% of the top fine-tuned result. As analyzed in the Impacts of Architecture section, this robust transferability likely stems from BIOT's STFT-based patch encoding, which explicitly captures the frequency-domain characteristics inherently required for emotion decoding.

For sleep staging on ISRUC, BIOT achieves superior performance due to extensive pretraining on large-scale clinical datasets, enabling effective encoding of sleep-stage dynamics. EEGPT also shows strong generalization on this task, achieving both the best mean performance and the best hyperparameter-optimized result despite not being pretrained on sleep datasets. As discussed previously, its linear patch projection and summary tokens preserve complete input information, enabling the model to effectively capture the distributed spatial coherence and long-range temporal context that sleep staging fundamentally requires.

Furthermore, we observe a notable cross-domain generalization capability among models pretrained exclusively on clinical corpora. Although the pretraining datasets for models such as CBraMod, BIOT, and NeuroGPT consist entirely of clinical recordings, they still demonstrate competitive generalization capabilities across non-clinical downstream tasks. This suggests that large-scale clinical EEG data contains fundamental neural patterns that can transfer effectively to broader cognitive and BCI applications.

4) Impacts of Preprocessing

CBraMod exhibits large standard deviations on all tasks except Movement, indicating unstable transfer. Specifically, its standard deviation reaches 11.07% on the Stress task but drops to a mere 0.21% on the Movement classification task, representing a 50-fold variance gap. This is likely due to its fixed 19-channel configuration during pretraining. When downstream datasets use different channel layouts, CBraMod must adapt to unseen channel mappings, leading to high variance. In contrast, the Movement dataset uses a 20-channel array (F7, FP1, FP2, F8, F3, FZ, F4, C3, CZ, P8, P7, PZ, P4, T3, P3, O1, O2, C4, T4, A2) that is highly similar to CBraMod's pretrained montage, and CBraMod correspondingly achieves its lowest standard deviation on this task. This outcome suggests strong sensitivity to channel mismatch. We hypothesize that pretraining with more diverse channel configurations would improve CBraMod's robustness across heterogeneous downstream tasks.

To address channel configuration mismatches in downstream tasks, the evaluated models adopt distinct strategies, including restricting inputs to the exact intersection of available channels, projecting channels via convolution, or directly using all channels. Specifically, LaBraM, NeuroLM, and EEGPT employ fixed, pretrained spatial embedding tables that support a specific vocabulary of 136, 139, and 62 standard electrodes, respectively. When applied to a new downstream task, these models must restrict the input data strictly to the intersection between the downstream dataset's channel configuration and their predefined embedding tables, discarding any downstream electrodes that lack a corresponding pretrained embedding. Because BIOT expects 16 channels, it utilizes a channelwise linear projection via a 1x1 Conv1d operation to align downstream inputs to the required shape. Conversely, Cbramod and ST directly process all available channels without structural alignment. In contrast, NeuroGPT employs a unique approach. Downstream EEG data are linearly mapped to the 22 channel NeuroGPT space using a mapping matrix based on a forward model. This matrix is computed from the fsaverage template utilizing a regularized pseudoinverse. This anatomically informed remapping explicitly models spatial relationships between electrodes.

This anatomically informed remapping explicitly models spatial relationships between electrodes. Despite not using a discrete codebook, NeuroGPT achieves relatively low standard deviation in 4 out of 5 tasks, suggesting that principled channel alignment can substantially improve stability under channel distribution shift.

VIII. Outstanding Questions

- What modeling and pretraining paradigms can balance the expanding parameter scales of LEMs with the scarcity of EEG data? While EEG data is more accessible than invasive neural recordings, its volume remains substantially smaller than the hyperscale corpora used in NLP and Computer Vision. As current models grow in complexity and depth, the demand for high-quality training data increases. Consequently, a critical challenge lies in developing data-efficient or sample-efficient strategies that can extract richer features from limited archives, thereby overcoming the bottleneck of data scarcity.
- Do conventional scaling laws apply to large-scale EEG models? Unlike language and vision foundation models, LEMs do not exhibit consistent power-law scaling with model size. We identify two interrelated explanations: (1) data scarcity limits exploration of computational regimes where such relationships emerge, and (2) task-specific architecture constraints prevent unified designs from generalizing across diverse EEG applications. Since different cognitive processes exhibit distinct temporal and spectral signatures, architectures optimized for one task often fail on others. This suggests that task-agnostic scaling may be fundamentally limited in neural signal processing. Can LEMs overcome these constraints through data augmentation, or do architectural requirements inherently limit conventional scaling laws?
- What architectural designs can overcome the limitations of implicit and explicit channel constraints? Despite varying architectural designs, most existing Large EEG Models impose implicit or explicit spatial constraints, compelling input signals to align with a fixed set of learnable embeddings or a pre-determined latent channel dimension. A key challenge is designing channel feature extraction methods that fully utilize the complete channel information available in diverse datasets.
- How can LEMs address current performance inconsistencies to achieve truly universal generalization? As evidenced by the results in this paper, while current models are applicable to multiple tasks, no single architecture consistently maintains superior performance across the full spectrum of downstream scenarios. This variability in cross-task efficacy suggests that current "Foundation Models" have not yet achieved the robustness required to serve as a universal backbone for BCI, especially in tasks related to emotion classification.
- How can LEMs evolve from predictive models to explanatory tools that drive neuroscientific discovery? The current opacity of learned features limits their scientific value. By aligning model representations with physiological evidence, we can overcome the "black box" limitation. This interpretability is the key to unlocking the true potential of NeuroAI, allowing large models to act as hypothesis generators that elucidate underlying neural mechanisms and facilitate broader applications in fundamental neuroscience.

IX. Conclusion

Our benchmarking results reveal several key factors governing performance across downstream EEG tasks. First, architectural design choices—particularly in patch encoding and embedding—critically influence task-specific performance. For tasks requiring extended

temporal context and multi-region coordination (e.g., sleep staging, emotion classification), linear projection-based tokenization outperforms more complex convolutional patch encoders. Conversely, tasks characterized by short temporal windows and localized channel-level patterns (e.g., ERN detection) benefit from expressive convolutional embeddings. Models incorporating frequency-domain features during patch encoding consistently achieve stronger linear probing performance, indicating that this design choice enhances feature separability for shallow classifiers.

Second, pretraining strategy fundamentally affects model robustness and generalization. Codebook-based pretraining (e.g., LaBraM) yields more stable feature representations and exhibits reduced sensitivity to hyperparameter variation. LLM-style EEG models demonstrate competitive performance with fewer fine-tuning epochs, though their full capacity may require larger-scale datasets than currently available.

Third, channel alignment strategies represent a critical preprocessing consideration. Models pretrained on limited channel configurations (e.g., fixed 19-channel montages) exhibit reduced transfer stability, while standardized channel conversion via professional EEG toolchains improves cross-dataset robustness. Finally, diverse pretraining corpora enhance performance on corresponding downstream tasks, suggesting that task-aligned pretraining data composition is a key design principle for foundation models in EEG analysis.

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APPENDIX A

TABLE XI
BACKBONE PARAMETER BREAKDOWN OF LABRAM ON SEED VII TASK

Component	Parameters
Temporal Convolution (1D)	576
CLS Token	200
Spatial Embedding	25,800
Temporal Embedding	6,400
Transformer Encoder (12 Layers)	5,789,760
Backbone Total	**5,822,736**

TABLE XII
BACKBONE PARAMETER BREAKDOWN OF CBRAMOD ON SEED VII TASK

Component	Parameters
Time Domain Pathway (1D Convolution)	5,200
Frequency Domain Pathway (FFT and FC)	20,400
ACPE Positional Encoding	26,800
Criss Cross Transformer (12 Layers)	4,831,200
Backbone Total	**4,883,600**

TABLE XIII
BACKBONE PARAMETER BREAKDOWN OF BIOT ON SEED VII TASK

Component	Parameters
Channel Adaptation	1,008
Patch Embedding	26,112
Channel Embedding	4,096
Transformer Encoder	3,155,968
Backbone Total	**3,187,184**

TABLE XIV
BACKBONE PARAMETER BREAKDOWN OF EEGPT ON SEED VII TASK

Component	Parameters
Patch Embedding	33,280
Channel Embedding	31,744
Summary Tokens	2,048
Transformer Encoder	25,219,072
Backbone Total	**25,286,144**

TABLE XV
BACKBONE PARAMETER BREAKDOWN OF NEUROGPT ON SEED VII TASK

Component	Parameters
Patch Embedding (CNN)	38,000
Transformer Encoder	118,320
Backbone Total	**156,320**

TABLE XVI
BACKBONE PARAMETER BREAKDOWN OF NEUROLM ON SEED VII TASK

Component	Parameters
VQ Tokenizer Temporal Convolution	309,888
VQ Tokenizer Positional and Temporal Embeddings	245,760
VQ Tokenizer Transformer Encoder	84,953,088
VQ Tokenizer Layer Norm	1,536
GPT 2 Decoder Token Embedding	38,633,472
GPT 2 Decoder Positional Embedding	786,432
GPT 2 Decoder Transformer	85,054,464
EEG Channel Positional Embedding	196,608
Backbone Total	**210,181,248**

TABLE XVII
BACKBONE PARAMETER BREAKDOWN OF ST TRANSFORMER ON SEED VII TASK

Component	Parameters
Channel Attention Block	14,090
Spatial Temporal Patch Embedding	305,728
Transformer Encoder (4 Layers)	3,159,040
Backbone Total	**3,478,858**